

CS-114 Fundamentals of Programming

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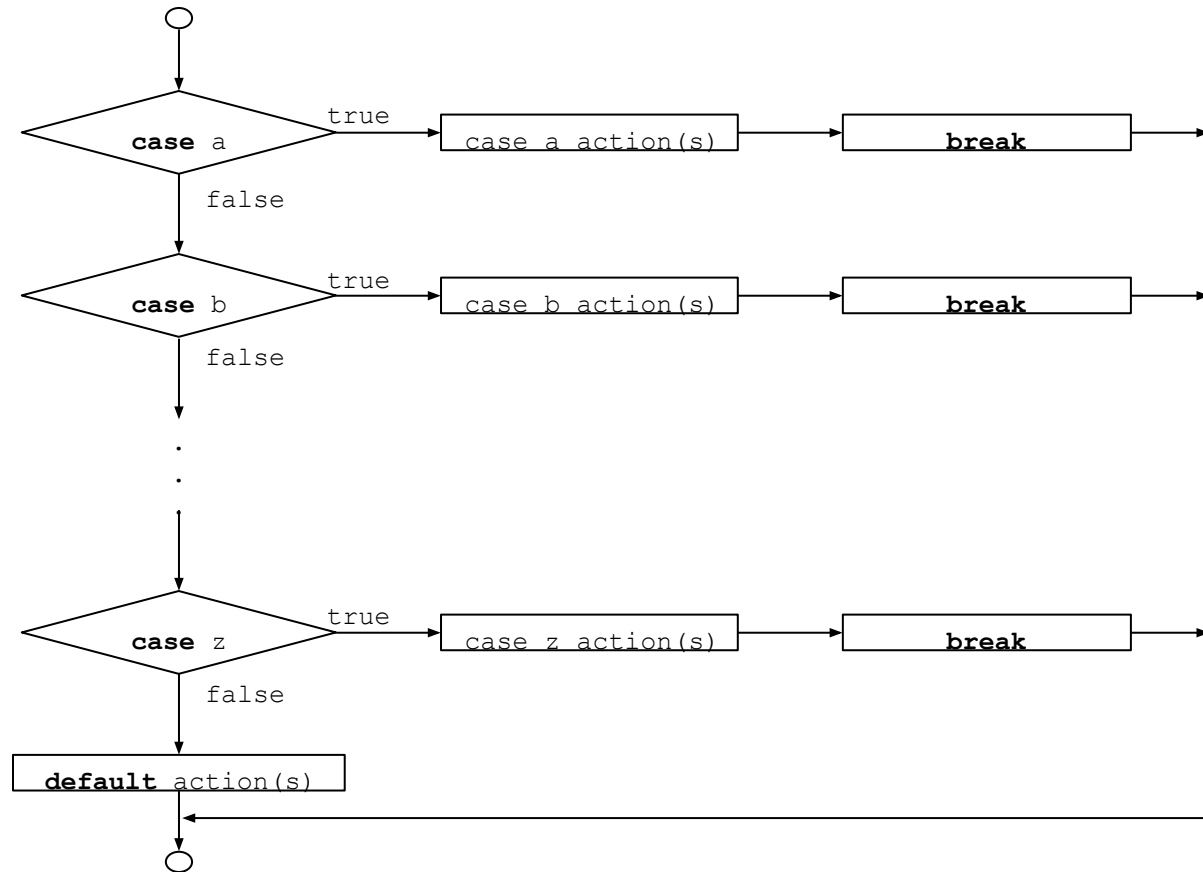
'switch' Multiple-Selection Structure

□ switch

- Test variable for multiple values
- Series of **case** labels and optional **default** case

```
switch ( variable ) {  
    case value1:           // taken if variable == value1  
        statements  
        break;             // necessary to exit switch  
  
    case value2:  
    case value3:           // taken if variable == value2 or == value3  
        statements  
        break;  
  
    default:               // taken if variable matches no other cases  
        statements  
        break;  
}
```

switch Multiple-Selection Structure



A program using switch statement that allows a user to choose an item from a menu so that the user's choice results in a unique message being printed out.

```
#include<iostream.h>

void main(void)
{
    int choice;

    cout<<"Choose one of the following"<<endl;
    cout<<"1 : Good Choice"<<endl;
    cout<<"2 : Have a nice day."<<endl;
    cout<<"3 : Here you go."<<endl;
    cout<<"4 : Cheers!"<<endl;

    cin>>choice;

    switch(choice)
    {
        case 1:
            cout<<"Good Choice."<<endl;
            break;
        case 2:
            cout<<"Have a nice day."<<endl;
            break;
        case 3:
            cout<<"Here you go."<<endl;
            break;
        case 4:
            cout<<"Cheers!"<<endl;
            break;
    }
}
```

switch Multiple-Selection Structure

□ Example 'switch'

- Program to read grades (A-F)
- Counts and display number of each grade entered

```
1 // Fig. 2.22: fig02_22.cpp
2 // Counting letter grades.
3 #include <iostream>
4
5 using std::cout;
6 using std::cin;
7 using std::endl;
8
9 // function main begins program execution
10 int main()
11 {
12     char grade;
13     int aCount = 0; // number of As
14     int bCount = 0; // number of Bs
15     int cCount = 0; // number of Cs
16     int dCount = 0; // number of Ds
17     int fCount = 0; // number of Fs
18
19     cout << "Enter the letter grades." << endl
20          << "Enter the EOF character to end input." << endl;
21     cin>> grade;
```

```
22 // loop until user types end-of-file key sequence
23 while (grade != EOF ) {
24
25     // determine which grade was input
26     switch ( grade ) { // switch structure nested in while
27
28         case 'A':        // grade was uppercase A
29         case 'a':        // or lowercase a
30             ++aCount;    // increment aCount
31             break;       // necessary to exit switch
32
33         case 'B':        // grade was uppercase B
34         case 'b':        // or lowercase b
35             ++bCount;    // increment bCount
36             break;       // exit switch
37
38         case 'C':        // grade was uppercase C
39         case 'c':        // or lowercase c
40             ++cCount;    // increment cCount
41             break;       // exit switch
42
```

```

22 // loop until user types end-of-file key sequence
23 while ( ( grade = cin.get() ) != EOF ) {
24
25     // determine which grade was input
26     switch ( grade ) { // switch structure
27
28         case 'A': // grade was uppercase A
29         case 'a': // or lowercase a
30             ++aCount; // increment aCount
31             break; // necessary to exit switch
32
33         case 'B': //
34         case 'b': //
35             ++bCount; // increment bCount
36             break; // exit switch
37
38         case 'C': // grade was uppercase C
39         case 'c': // or lowercase c
40             ++cCount; // increment cCount
41             break; // exit switch
42

```

break causes **switch** to end and the program continues with the first statement after the **switch** structure.

cin.get() uses dot notation (explained chapter 6). This function gets 1 character from the keyboard, and it is assigned to **grade**.

cin.get() returns EOF (end-of-file) after the EOF character is input, to indicate the end of data. EOF may be ctrl-d or ctrl-z, depending on your OS.

The value of this statement is the same as the value returned by **cin.get()**.

Compares **grade** (an **int**) to the numerical representations of **A** and **a**.


```

43     case 'D':           // grade was uppercase D
44     case 'd':           // or lowercase d
45         ++dCount;       // increment dCount
46         break;          // exit switch
47
48     case 'F':           // grade was uppercase F
49     case 'f':           // or lowercase f
50         ++fCount;       // increment fCount
51         break;          // exit switch
52
53     case '\n':          // ignore newlines,
54     case '\t':          // tabs,
55     case ' ':           // and spaces in input
56         break;          // exit switch
57
58     default:            // catch all other characters
59         cout << "Incorrect letter grade entered."
60             << " Enter a new grade." << endl;
61         break;          // optional; will exit switch anyway
62
63 } // end switch
64
65 } // end while
66

```

Notice the **default** statement, which catches all other cases.

```
67 // output summary of results
68 cout << "\n\nTotals for each letter grade are:"
69     << "\nA: " << aCount // display number of A grades
70     << "\nB: " << bCount // display number of B grades
71     << "\nC: " << cCount // display number of C grades
72     << "\nD: " << dCount // display number of D grades
73     << "\nF: " << fCount // display number of F grades
74     << endl;
75
76 return 0; // indicate successful termination
77
78 } // end function main
```

Enter the letter grades.
Enter the EOF character to end input.

a
B
c
C
A
d
f
C
E
Incorrect letter grade entered. Enter a new grade.
D
A
b
^Z

Totals for each letter grade are:
A: 3
B: 2
C: 3
D: 2
F: 1